

The influence of the Ukraine-Russia conflict on renewable and fossil energy price cycles

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ARTICLE INFO

JEL:
C50
F51
Q41

Keywords:

Renewable energy
Fossil energy
MS-BVAR
Ukraine-Russia conflict

ABSTRACT

This study investigates the impact of the Ukraine-Russia conflict on energy cycles by extending a business cycle-based approach in the energy sector. We employ the Markov Switching Bayesian Vector Autoregressive (MS-BVAR) model to derive filtered probabilities that represent energy cycles. By utilizing the Russia Economic Policy Uncertainty Index and dummy variables derived from Google search data as conflict indicators, we conduct regression analysis and examine the generalized impulse response function to assess the effects of the conflict on energy cycles. Our research reveals notable differences in renewable and fossil energy price cycles. The renewable price cycle exhibits greater persistence, indicating a limited response to the conflict shock. In contrast, the fossil energy price cycle exhibits more significant and enduring adjustments in high- and low-volatility regimes. These findings highlight the heterogeneous impacts of the Ukraine-Russia conflict on the fossil energy market, indicating potential disruptions in supply chains and market sensitivities. Notably, fossil energy prices demonstrate higher responsiveness to the Ukraine-Russia conflict across market conditions.

1. Introduction

The conflict between Ukraine and Russia has substantial implications for energy markets, including concerns about potential disruptions in supply routes and geopolitical tensions (Ferriani and Gazzani, 2023; De Villa, 2023; Saâdaoui and Jabeur, 2023; Goodell et al., 2023). The tensions between Russia and Ukraine can lead to trade wars and economic sanctions between Russia and the world, which can reduce global trade and investment and slow down economic growth. Political instability and conflict in crucial regions can also disrupt energy supplies and increase commodity prices, reducing consumer purchasing power (Goodell et al., 2023).

Despite its significance, there has been limited research to date that has examined the impact of the Ukraine-Russia tension on both fossil and renewable energy markets. This gap represents a significant void in the research literature. Therefore, our study takes a unique approach to comprehending the multifaceted influence of the Ukraine-Russia conflict, utilizing the Economic Policy Uncertainty (EPU) index in conjunction with a dummy variable based on Google Trends data in innovative ways for our analysis. Unlike traditional geopolitical risk index, it captures the geopolitical dimensions and provides a comprehensive view of economic uncertainties, including monetary and fiscal

policy changes, regulatory fluctuations, and broader economic factors. This approach strengthens the study by enabling a deeper understanding of the conflict's long-term economic repercussions, its relative significance in comparison to other economic uncertainties, and its multifaceted impact across various sectors, making it a more persuasive and versatile tool for analyzing the intricate relationship between geopolitics, economic policy, and energy markets. This specificity ensures that the analysis is finely tuned to the energy market consequences of the conflict (Pata et al., 2023). Our motivation extends to several compelling aspects. Firstly, the Ukraine-Russia conflict remains an ongoing and dynamic geopolitical event with widespread implications, making it an ideal subject for in-depth examination. Secondly, by incorporating a dummy variable based on Google Trends data, which gauges the public's real-time awareness of the phrase "Ukraine and Russia Tensions," we aim to complement traditional data sources with insights from public perception. Furthermore, our study ventures into the realm of energy cycles, similar to business cycles but specific to the energy sector. As geopolitical events substantially shape energy markets, our research aims to shed light on the intricate relationship between the conflict and these energy cycles. In doing so, we enhance our research methods and contribute to a deeper understanding of the complex relationship between geopolitical events and the energy landscape, providing valuable

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<https://doi.org/10.1016/j.eneeco.2023.107218>

Received 18 June 2023; Received in revised form 27 September 2023; Accepted 20 November 2023

Available online 30 November 2023

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insights for researchers, policymakers, and industry stakeholders.

Motivated by this research gap and supported by evidence of significant energy price fluctuations linked to Ukraine and Russia tensions (Zhou et al., 2023), we investigate the effects of these tensions on both fossil and renewable energy price cycles. The primary underlying inquiry we aim to tackle revolves around whether the energy markets and well-informed traders active within these markets have, over an extended period, subtly conveyed indications regarding the ongoing evolution of energy supply risks that are now becoming more evident.

Considering the data from 2010 to 2020, which covers the onset of the Russia-Ukraine conflict in 2014 and extends to the Russia-Ukraine war in 2022, the study's analysis using the Markov-Switching Bayesian Vector Autoregressive (MS-BVAR) model, regression model, and impulse response function analysis has unveiled several significant findings. Firstly, it highlights distinct disparities between renewable and fossil energy prices, with renewables displaying a greater degree of persistence and volatility regime shifts over time. Notably, renewable energy prices predominantly stayed in a low volatility regime from 2010 to 2018, signaling a market downturn, but shifted to higher fluctuations from 2019 to 2022, indicative of an upturn. In contrast, fossil energy prices exhibited less persistence and demonstrated significant regime switching following the Ukraine-Russia conflict escalation. Secondly, empirical evidence from regression results reveals that the Ukraine-Russia conflict has a more pronounced impact on fossil energy prices compared to renewable energy prices, potentially due to supply chain disruptions, geopolitical tensions, and market sensitivities. Lastly, the generalized impulse response function analysis reveals that the impact of the conflict on renewable energy prices is relatively modest and short-lived, while fossil energy prices experience a sustained and more pronounced response, particularly in the low volatility regime, with coal prices showing the highest responsiveness. These findings provide valuable insights into the divergent impacts of the conflict on renewable and fossil energy markets. Our findings raise concerns regarding the concentration risks inherent in both renewable and fossil fuel markets, encompassing aspects such as supply sources and supply chains. These observations highlight the urgency of implementing policy measures aimed at reducing such dependencies. Furthermore, our results emphasize the need for market participants to be aware of the vulnerabilities of energy supplies in the context of the Ukraine-Russia conflict, even before policy responses are enacted.

The conflict and war between Russia and Ukraine have significantly influenced global energy prices, primarily due to Russia's position as the world's third-largest oil producer and exporter, ranking just behind the United States and Saudi Arabia (Yag'ya and Yagya, 2021). As a crucial exporter of commodities like crude oil, natural gas, and coal, Russia's actions have reverberated throughout the energy markets on a global scale. Despite the invasion of Ukraine by Russia on February 24, 2022, the immediate impact on the world oil supply was not a decrease. However, global oil and natural gas prices experienced a notable surge (International Energy Agency, 2021). In addition to military actions, Western countries and their allies, including the United States, the European Union, the United Kingdom, Australia, Canada, Germany, and Japan, have responded by imposing embargoes and trade restrictions on Russian goods and commodities. These accompanying sanctions have directly impacted global energy prices, as they have constrained the demand for fossil fuels (Mahlstein et al., 2022).

Recognizing the need for a strategic response, many countries have devised measures to reduce dependence on Russian energy. One of the proposed strategies involves accelerating the development of renewable energy sources (Mohammed et al., 2023). Given the intricate geopolitical landscape and the interconnectedness of global energy markets, the conflicts and wars between Ukraine and Russia have far-reaching influences on energy prices. The ongoing volatility, coupled with the

imposition of sanctions, has disrupted supply and demand dynamics, leading to significant fluctuations in global energy markets (Li and Li (2022)). Consequently, diversifying the energy mix and promoting the adoption of renewable energy sources have emerged as crucial priorities for countries seeking to mitigate the impact of such conflicts on energy prices and enhance their overall energy security (Karkowska and Urjasz, 2023a).

Renewable energy, with its remarkable capacity to harness the power of nature through solar, wind, and hydropower, emerges as a beacon of hope amidst the challenges posed by reduced energy supply due to embargoes (Umar et al., 2022). Beyond its practical solution to supply constraints, renewable energy holds the transformative potential to address pressing environmental issues, offering a sustainable alternative to the destructive path of fossil fuels (Czech and Wielechowski, 2021). Notably, the advantages of renewable energy extend beyond the environmental realm. Harnessing local resources' power ensures energy security, reducing reliance on foreign sources and mitigating the vulnerabilities associated with geopolitical conflicts (Bulavskaya and Reynès, 2018; Musah et al., 2020). As the urgency to embrace renewable energy intensifies amidst ongoing conflicts, global leaders have taken up the mantle of change, spearheading the charge towards a sustainable future (Pototschnig et al., 2022).

Our study seeks to shed light on the conflict's intricate influence on fossil and renewable energy cycles. To evaluate this impact, we suggest using the EPU developed by Baker et al. (2016) and augmenting it with a dummy variable derived from Google Trends data, enhancing the reliability and informativeness of our analysis. The EPU Index is constructed using diverse economic-related media data from various offline and online sources, including news media sentiment, tax code changes, trade policy shifts, monetary policy actions, and fiscal policy developments (Baker et al., 2016). It provides us with a valuable means to monitor changes in economic policy uncertainty, a variable with significant implications for investment decisions, economic forecasts, and overall economic stability. Significantly, there has been a notable increase in EPU, signifying ambiguity and uncertainty in economic, social, and international relations (Gu et al., 2021; Pata et al., 2023). This rise in EPU is a global phenomenon, driven by major recent events, including the 2008 global financial crisis (Fang et al., 2018), the far-reaching impact of the COVID-19 pandemic (Dou et al., 2022), the US-China trade war (Chen et al., 2023), and the protracted Russia-Ukraine conflict (Pata et al., 2023; Shen and Hong, 2023). Furthermore, numerous research studies have explored the impact of EPU on both renewable and fossil energies (Fang et al., 2018; Dash and Maitra, 2021; Ringim et al., 2022; Wang et al., 2023). This emphasizes the EPU index's suitability as a comprehensive representative measure for assessing the profound impacts of the Ukraine-Russia conflict on renewable and fossil energy price cycles, especially within macroeconomic contexts.

However, given that the EPU index encompasses factors beyond the specific Russia-Ukraine conflict, it is necessary to isolate the impact of this conflict specifically. To achieve this, a dummy variable is introduced, which distinguishes periods of conflict from periods of peace. The dummy variable takes on a value of 1 to indicate the presence of the conflict and 0 to indicate its absence. By multiplying the EPU index with the dummy variable, we obtain a refined indicator that specifically captures the influence of the Russia-Ukraine conflict. Furthermore, it is important to note that the impact of the conflict on renewable and fossil energies, particularly in terms of price cycles, remains a complex and intriguing area of investigation. The limited existing research in this domain presents both challenges and opportunities for further exploration.

This study aims to address several key aspects that have not been adequately explored in prior research. Firstly, this study aims to fill a significant gap in the existing literature by conducting a comprehensive

analysis of the influence of the Ukraine-Russia conflict on the price cycles of both renewable and fossil energy sources. Unlike previous studies that overlooked the relationship between the conflict and energy price cycles, this research takes a holistic approach, considering energy prices as a whole. By examining their aggregate behavior, a more nuanced understanding of how the conflict impacts the energy market can be obtained.

Secondly, to quantify price cycles in both renewable and fossil energy markets, this study employs the MS-BVAR model. The decision to use the MSBVAR model is driven by the need to estimate energy cycles that extend beyond the capabilities of traditional threshold and Smooth Transition VAR models. While the threshold VAR model relies on pre-determined thresholds and may struggle to capture the intricate complexities of energy price cycles, particularly during periods of geopolitical turmoil, and Smooth Transition VAR models may lack the flexibility to model dynamic regime shifts effectively, the MS-BVAR model excels in autonomously identifying different states or regimes within the data. This inherent adaptability is particularly advantageous when examining energy cycles influenced by the Ukraine-Russia conflict, where regime shifts can be unpredictable and dynamic. The MS-BVAR model's proficiency in capturing nonlinear dynamics and regime transitions makes it a potent tool for estimating energy cycles in a data-driven and comprehensive manner. In particular, the MS-BVAR framework offers the advantage of not requiring the specification of a specific variable that triggers the transition between regimes, whether abrupt (threshold) or smooth. Furthermore, the smooth transition model necessitates the estimation of a greater number of parameters, which can potentially lead to increased estimation errors (Sun et al., 2022).

Lastly, the study investigates the impulse response of various energy prices within renewable and fossil energy markets to a shock caused by the Ukraine-Russia conflict. This analysis goes beyond previous research by considering different regimes within the price cycles, such as low and high volatility periods (market downturn and upturn, respectively). By examining the responses under varying conditions, the study provides a detailed illustration of the dynamics and sensitivities of energy prices to the conflict. These findings have important implications for informed decision-making at the macro-level within the energy sector, enabling policymakers and industry participants to better understand the implications of the conflict and devise effective strategies in response.

The remainder of this paper is organized as follows: Section 2 presents the literature review. Section 3 presents the methodology and data. Section 4 discusses the results. Section 5 provides the empirical results. Finally, Section 5 includes the conclusions and policy implications.

2. Literature review

2.1. Cyclical energy prices

The concept of the business cycle, defined by Burns and Mitchell (1946), refers to the recurring pattern of economic fluctuations that affect all sectors of the economy over time. These fluctuations are influenced by various economic events occurring at different intervals. While each cycle within the business cycle does not have a fixed duration, its length depends on the severity of the economic conditions. The business cycle consists of four phases: expansion, peak, recession, and trough.

Undoubtedly, cycles form an essential characteristic of energy price dynamics, representing recurrent patterns of fluctuation (Khan et al., 2022). Energy prices undergo prolonged periods marked by gradual appreciation, interspersed with abrupt downward corrections that disrupt the trend (Deaton and Laroque, 1992). Building upon Kondratieff's seminal work in 1925 (Kondratieff, 1925), subsequent economists have recognized the presence of both short and long-term output cycles within these cyclical fluctuations. Subsequently, economists

recognized that these business cycles are correlated with cycles in other macroeconomic variables, including commodity prices (Ott and Tatom, 1986).

In alignment with the business-cycle literature originating from Burns and Mitchell (1946), energy price booms are characterized by substantial price increases, while busts denote periods of price decreases. Notably, Reinhart and Wickham (1994) demonstrated that energy prices display super cycles and synchronized movement during these cycles. The duration, magnitude, and frequency of energy price cycles vary depending on the specific energy source and prevailing market conditions (Agnello et al., 2020). Empirical evidence reveals cyclical patterns in oil prices, wherein periods of elevated prices are propelled by factors such as geopolitical tensions, supply disruptions, or heightened demand, followed by phases of lower prices driven by factors like improved supply, economic slowdowns, or shifts in energy policies (Gil-Alana and Gupta, 2014).

Cashin and McDermott (2002) conducted a comprehensive analysis of commodity price cycles, revealing an interesting asymmetry: price contractions typically endure longer than price booms. Furthermore, they observed that the probability of a contraction or boom in prices ending remains independent of the duration of that particular phase. Expanding on this research, Erdem and Ünalmış (2016) employed diverse filtering techniques to characterize super-cycles in oil prices dating back to 1861. Their findings indicated that starting from the 1990s, emerging economies' rapid growth and asset price booms in advanced countries stimulated both oil demand and prices.

In this study, our primary objective is to apply the business cycle concept specifically to fossil and renewable energy prices. However, it is essential to acknowledge that our research's limited 10-year time frame may pose challenges in observing all four phases of the energy price cycle. Therefore, our focus will be on identifying and analyzing the phases characterized by upward movement or high volatility in energy prices and the phases of downward movement or low volatility. These identified phases will form the fundamental assumptions underlying our research. Furthermore, in studying the "short-term" cycles, previous research has predominantly relied on relatively higher-frequency data, such as monthly data, to capture the trends in energy prices and identify boom-bust episodes. By adopting this approach, we aim to gain insights into the cyclical behavior of energy prices and understand how they are influenced by factors such as the ongoing Ukraine-Russia conflict. Through this analysis, we intend to contribute to a more profound understanding of the dynamics within the energy market and provide valuable information for policymakers, investors, and stakeholders in the energy sector.

2.2. Energy, conflict, and war

Demand, supply, and investor concerns are widely recognized as the primary factors influencing energy, as discussed by Shafiee and Topal (2010). The energy demand is intricately linked to global economic conditions, and periods of economic recession, such as the financial crisis in 2007–2008 or the ongoing COVID-19 pandemic, introduce uncertainties that significantly impact energy demand, as highlighted by Yu et al. (2022). These uncertainties, compounded by lockdown measures and reduced travel, have notably resulted in a significant downturn in demand, consequently leading to a decrease in energy prices. Furthermore, the demand side of the energy market encompasses a spectrum of factors beyond economic conditions, including technological advancements, energy efficiency initiatives, government policies, and evolving consumer preferences. The relationship of these variables contributes to the complex and dynamic nature of energy demand, making it a multifaceted facet of the broader energy landscape (Berndt and Wood, 1975).

Indeed, it is crucial to recognize that energy prices can be influenced

by demand factors and supply dynamics. A notable example of this is the Yom Kippur War, where the conflict between Arab and Israeli nations resulted in a significant increase in oil prices due to a decrease in supply. Arab-majority nations forming the Organization of the Petroleum Exporting Countries (OPEC) utilized oil as a political tool to exert pressure on Israel, leading to fluctuations and a surge in energy prices, as explained by Schumacher (1985). Numerous other events throughout history have also caused volatility in energy prices. For instance, the Persian Gulf War in 1990 resulted in a temporary disruption in oil exports, while the Venezuelan protests in 2001 and 2002 impacted oil production. The Second Persian Gulf War in 2003, the Syrian civil war and protests, the Libyan crisis in 2011, and the Russo-Ukrainian War in 2022, as highlighted by Zhou et al. (2023), have all had significant implications for energy prices. These examples serve as reminders that energy prices often experience cyclical fluctuations and are heavily influenced by major events, such as wars, which can disrupt supply dynamics and have a profound impact on the energy market.

In recent times, extensive discussions among academics and researchers have revolved around the Ukraine-Russian war's influence on the energy sector (Benton et al., 2022). The conflicts and hostilities between Ukraine and Russia have left a profound and enduring mark on global energy prices, as emphasized in Zhou et al.'s study Zhou et al. (2023). The subsequent imposition of embargoes on Russia has triggered economic retaliation between Russia and other nations, increasing volatility within the crude oil and natural gas markets (Dreger et al., 2016). The pivotal events in 2014 and 2022 when Russia invaded Ukraine were of particular significance, meticulously documented by Ringim et al. (2022). The ramifications of this conflict have rippled throughout the energy market, shedding light on the intricate relationship between geopolitical tensions and fluctuations in energy prices.

The recent geopolitical tremors stemming from the Ukrainian crisis have reignited concerns regarding the reliance on conventional energy sources. Geopolitical tensions between Ukraine and Russia exacerbate the risk of disrupting the global flow of fossil fuels. Such tensions can pressure nations to reduce their dependence on fossil fuels and explore greener alternatives. The current Ukraine crisis and recent events may serve as a catalyst for countries to lessen their reliance on fossil fuels and diversify their energy portfolios (Cheikh and Zaid, 2023; Mohammed et al., 2023).

The ongoing tension between Ukraine and Russia has notably impacted the natural gas market in Europe (Goodell et al., 2023). This is primarily due to the Nord Stream pipeline linking Russia to Germany via the Baltic Sea. On one hand, it has raised energy security concerns among some European nations, who worry about overreliance on Russian gas and potential disruptions due to geopolitical tensions. Consequently, the natural gas market in Europe is not solely influenced by the direct consequences of Ukraine-Russia tensions but also by the broader implications of projects like Nord Stream. These intricate dynamics underscore the interconnected nature of energy markets and their sensitivity to geopolitical developments.

Furthermore, the ongoing tension between Ukraine and Russia can potentially impact coal prices (Nerlinger and Utz, 2022). High natural gas prices had previously pushed Europe towards coal; however, there was limited availability of coal on the global market due to Western sanctions, making it challenging to trade Russian coal, which is the third-largest coal exporter.

2.3. EPU and energy prices

Numerous studies have extensively investigated the profound impact of economic and policy uncertainty on energy prices. Economic and policy uncertainty can influence energy prices through various vital mechanisms.

Firstly, economic uncertainty affects investment decisions within the energy industry. During heightened uncertainty, companies often resist investing in new projects or expanding production capacities. Consequently, this cautious approach leads to a reduction in energy supply, exerting upward pressure on prices (Antonakakis et al., 2014; Qin et al., 2020; Dash and Maitra, 2021; Gu et al., 2021).

Secondly, economic uncertainty significantly influences consumer and business spending, subsequently affecting the oil demand. In times of elevated uncertainty, such as economic recessions or financial crises, overall economic activity tends to decelerate, reducing oil demand and potentially lowering prices (Kang and Ratti, 2013; Dash and Maitra, 2021). Reducing consumer confidence due to uncertainties surrounding financial well-being, job security, and future income prospects plays a pivotal role in shaping spending decisions. When faced with uncertainty, individuals exhibit caution by postponing major purchases, cutting back on discretionary spending, and increasing savings as a precautionary measure. This decline in consumer spending within the energy industry exerts downward pressure on energy prices (Dees and Brinca, 2013).

Thirdly, it is essential to consider the substantial influence that geopolitical tensions can exert on energy prices. The uncertainty surrounding policies triggered by geopolitical events and conflicts can disrupt the supply chains and energy production. Unforeseeable policy actions, trade disputes, or regional conflicts often raise concerns about potential disruptions in the supply, thereby causing fluctuations in energy prices. A pertinent illustration of this phenomenon can be found in the Russian invasion of Ukraine, which reverberated throughout global financial markets and affected the renewable energy sector. Investors, recognizing the imperative for alternative energy sources during geopolitical instability, have increasingly turned their attention to the renewable energy sector (Umar et al., 2022). Furthermore, Goodell et al. (2023) have provided compelling insights into the profound impact of geopolitical events on energy pricing, with a particular focus on natural gas. Their research underscores the pivotal role played by the Russia-Ukraine conflict in shaping the dynamics of energy pricing. Notably, their study reveals an intensified correlation between announcements related to the Nord Stream pipeline and the subsequent market responses within the Dutch TTF natural gas market. This highlights the energy market's remarkable ability to swiftly incorporate and react to geopolitical risks within its pricing mechanisms. To illustrate the consequences of such disruptions, consider the impact on natural gas supplies, which leads to heightened demand for expensive alternative fuels and increased uncertainty surrounding energy infrastructure and financing for renewable energy projects (Saâdaoui and Jabeur, 2023).

Lastly, economic and policy uncertainty influences energy prices through speculative trading in financial markets. Traders and investors actively try to anticipate the ramifications of uncertainty on supply and demand dynamics, thereby contributing to heightened volatility in price movements. A noteworthy illustration of this phenomenon is observed in the context of the war between Ukraine and Russia, which has resulted in a reduction in oil supply. Consequently, this reduction has driven up energy prices and generated higher returns (Karkowska and Urjasz, 2023b). Additionally, Liu et al. (2021) have elucidated that uncertainty can significantly impact investor expectations, leading to declines in energy-related stock returns. Within this context, it is common for investors to adopt a more cautious approach and choose to delay their investment decisions, aligning with the principles outlined in the fundamental options theory. Ultimately, such cautious behavior contributes to the reduction of stock returns.

The existing literature review highlights a notable gap in our understanding of the significant impacts of the Ukraine-Russia conflicts on fossil and clean energy markets. The relationship between these conflicts and energy markets remains insufficiently explored, particularly regarding its implications for investors and regulators. As a result, this

study seeks to address this research gap and contribute to the global literature by providing insights into this crucial area of investigation.

Its extensive adoption in evaluating the impact of political and economic uncertainty on the energy sector underscores its significance. Numerous studies have utilized the EPU index to uncover the intricate relationship between uncertainty and energy markets. One notable investigation by Gu et al. (2021) delved into the effects of EPU on the oil market, employing a Structural Vector Autoregressive (SVAR) model. Their findings revealed a pronounced negative effect, as pervasive uncertainty overshadowed the market, leading to hesitancy and ambiguity. In a parallel study, Qin et al. (2020) scrutinized the consequences of the trade war between the United States and China through the lens of the relationship between EPU and oil prices. Employing an Equilibrium model and Wavelet analysis, their investigation highlighted a distinct adverse influence of EPU on oil prices. The trade war initiated a surge in economic policy uncertainty, triggering heightened volatility and instability in oil markets. Notably, exploring EPU's correlation with fossil fuel prices, such as crude oil and natural gas, has encompassed Russia. Ringim et al. (2022) employed multivariate Dynamic Conditional Correlation Generalized Autoregressive Conditional Heteroscedasticity (DCC-GARCH) models to uncover a robust linkage between Russian EPU and crude oil and natural gas prices. They revealed that this relationship remained resilient despite various crises, including financial downturns, terrorist attacks, and conflicts between Russia and Ukraine.

3. Data and methodology

3.1. Data

This study analyzes the energy dynamics and response of renewable and fossil energy sectors during the Ukraine and Russia tensions, conflict, and war. The chosen analysis period spans from January 2010 to December 2022, encompassing the pre-war period and the intensification of tensions. By examining this timeframe, the study aims to capture the evolving energy landscape and the impact of geopolitical events on the energy markets. The decision to investigate the energy cycle during this period is driven by the recognition that the tensions and conflicts between Russia and Ukraine have gradually escalated over time. Before the outbreak of the war in 2014 and the heightened tensions in 2022, there were signs of increasing strain and apprehension between the two countries.

The data utilized in this study were obtained from reputable sources, including www.investing.com, Refinitiv Eikon, and the CEIC database. To represent the renewable energy sector, three prominent indexes were selected: the WilderHill Clean Energy Index (ECO), which includes stocks of companies involved in advancing cleaner energy and conservation; the European Renewable Energy Index (ERIX), which measures the performance of major renewable energy companies in Europe across

various categories such as wind, solar, biomass, and water energy; and the S&P Global Clean Energy Index (SPGC), which tracks the performance of companies engaged in clean energy-related businesses globally, encompassing both developed and emerging markets. These index choices have been widely adopted in previous studies as representative measures of the renewable energy sector (Mejdoub and Ghorbel, 2018; Paiva et al., 2018; Reboredo et al., 2019; Asl et al., 2021; Czech and Wielechowski, 2021; Hemrit and Benlagha, 2021; Liu et al., 2021). For the representation of fossil energy prices, three variables were used: Brent crude oil spot price (BOIL), Natural gas TTF spot price (NG), and Global coal price (CP). These variables have been widely employed as benchmarks in previous studies examining fossil energy markets (Paiva et al., 2018; Joo et al., 2020; Ghabri et al., 2021; Maneejuk et al., 2022).

To accurately assess the impact of the Ukraine-Russia conflict on energy markets, this study exclusively utilizes the Russia Economic Policy Uncertainty (REPU) index as a proxy for the conflict. The rationale behind this choice is rooted in the significant role played by Russia as the largest global producer and exporter of fossil energy, combined with the enduring nature of the conflict between Russia and Ukraine. It is worth noting that Ukraine's influence on world fossil prices is relatively limited, given its position as neither a major importer nor exporter of fossil energy. Additionally, the availability of Ukraine's Economic Policy Uncertainty data is notably scarce. Therefore, the study focuses solely on the REPU index, which provides a comprehensive measure to capture the multifaceted impacts of the Russia-Ukraine conflict on energy markets. However, since the REPU index encompasses factors beyond the specific Russia-Ukraine conflict, a carefully designed approach is implemented to isolate the impact of this conflict. A dummy variable is introduced to differentiate the periods of conflict from those of peace. The dummy variable takes on a value of 1 to indicate the presence of the conflict and 0 to indicate its absence. By multiplying the REPU index with the dummy variable, a suitable indicator is derived precisely capturing the Russia-Ukraine conflict's influence. To determine the periods of conflict, Google Trends data related to the keyword "Ukraine and Russia Tensions" is utilized as a reference search. A binary value of 1 is assigned when the Google Trends index exceeds zero, indicating the presence of a conflict, while 0 signifies its absence. This methodology enables the assessment of the specific time periods when the conflict between Ukraine and Russia occurred and its potential impact on the energy cycle.

Furthermore, the veracity of the data is ensured through validation against contemporaneous news reports. In addition, this study employs three control variables to avoid Omitted Variables, including the US dollar index (USD), Interest rate (INR) from 10-year United States treasury bills, and the Industrial Production Index (IPI). Before analysis, all of the data will be transformed into the form of a natural logarithm. Table 1 shows the data description and the variable.

Table 2 provides a comprehensive overview of the dataset by presenting descriptive statistics. Table 2 offers a comprehensive overview of

Table 1
Data description and variable.

Variable	Symbol	Definition	Source
Wilder Hill Clean Energy Index	ECO	Renewable Energy Index	Refinitiv Eikon
The European Renewable Energy Index	ERIX	Renewable Energy Index	investing
S&P Global Clean Energy Index	SPGC	Renewable Energy Index	investing
Brent crude oil spot price	BO	Fossil Energy price	CEIC database
Global Coal price	CP	Fossil Energy price	Refinitiv Eikon
Natural Gas TTF spot price	NG	Fossil Energy price	Refinitiv Eikon
Russia's Economic policy uncertainty index	REPU	Ukraine-Russia conflict	EPU Indices
Dummy	Dum	Ukraine-Russia conflict	Google Trend
Renewable Energy Cycle	REC	Renewable Energy Cycle	Estimation
Fossil Energy Cycle	FEC	Fossil Energy Cycle	Estimation
US Dollar index	USD	Control Variable	Investing
Interest rate	INR	Control Variable	investing
Industrial Production Index	IPI	Control Variable	Federal Reserve Economics data

Table 2
Descriptive statistics.

Variable	lnECO	lnERIX	lnSPGC	lnBO	lnNG	lnCP	lnREPU	lnUSD	lnINR	lnIPI
Mean	4.211	6.794	6.764	4.283	3.956	5.808	5.279	4.499	0.730	4.596
Median	4.071	6.798	6.640	4.275	3.917	5.783	5.324	4.533	0.795	4.603
Max	5.512	7.967	7.804	4.835	5.857	6.726	6.677	4.720	1.399	4.652
Min	3.642	5.645	6.112	3.271	2.140	4.795	3.600	4.290	−0.629	4.442
Std. Dev.	0.454	0.576	0.383	0.354	0.589	0.474	0.592	0.099	0.391	0.040
Skewness	1.019	0.095	0.879	−0.342	0.861	−0.040	−0.145	−0.265	−1.217	−1.145
Kurtosis	3.234	2.586	3.075	2.410	5.396	2.367	2.782	1.986	4.832	4.403

the dataset, presenting descriptive statistics. We applied log transformation to all variables. Log transformation is a commonly employed technique in econometric analysis, particularly with economic and financial data. It serves to stabilize variances and reduce the impact of extreme values. This transformation proves especially beneficial when dealing with time series data, which often exhibit heteroscedasticity and non-constant volatility, as noted in previous studies (Huang et al., 2021; de la Horra et al., 2022). Furthermore, log-transformed data often align more closely with the assumptions of regression models, facilitating the application of statistical tests and result interpretation. Additionally, it offers the advantage of making percentage changes directly interpretable as approximate elasticities. We acknowledge that the use of log transformations is a common practice in the field of price cycle analysis. Sun et al. (2022) suggested that log-transformed variables can be employed in the MS-VAR model when cointegration exists among the levels of the variables in question.

Among the variables, lnERIX has the highest mean value of 6.794, indicating that, on average, it has the highest performance among the renewable energy indexes. lnSPGC closely follows it with a mean of 6.764, lnCP with a mean of 5.808, and lnREPU with a mean of 5.279, respectively. In terms of the index values themselves, lnERIX records the highest value at 7.967, indicating strong performance in the European renewable energy sector. It is followed closely by lnSPGC with a value of 7.804. On the other hand, lnINR exhibits a low negative value of −0.629, suggesting a relatively weaker performance of the Indian Rupee exchange rate. The standard deviation, which measures the volatility of the variables, shows that lnREPU has the highest volatility with a value of 0.592, indicating greater price fluctuations in the Russian Economic Policy Uncertainty Index. lnERIX follows closely with a value of 0.576, indicating relatively high volatility in the European Renewable Energy Index. Conversely, lnIPI, lnUSD, and lnNG exhibit lower price volatilities with values of 0.040, 0.099, and 0.284, respectively. The skewness values indicate the asymmetry of the distributions. The negative skewness values for lnBO, lnCP, lnREPU, lnUSD, lnINR, and lnIPI indicate left-sided skewness, suggesting that these variables are skewed towards lower values. The kurtosis values measure the tailedness of the distributions. lnINR and lnECO show the highest kurtosis values of 4.832 and 3.234, respectively, indicating heavier tails in their distributions compared to a normal distribution.

3.2. Methodology

3.2.1. Markov switching Bayesian vector autoregressive model (MS-BVAR)

This study employs the Markov switching Bayesian Vector Autoregressive Model (MS-BVAR) to estimate the filtered probability, which serves as a proxy for energy cycles. Specifically, the MS-BVAR model is applied to model fossil and renewable energies separately, allowing for estimating two distinct cycles. Using the MS-BVAR model, we can capture the dynamic nature of energy markets and identify different regimes or states within each energy sector. This approach also enables us to examine the behavior and interactions of variables in each regime and gain a deeper understanding of the underlying factors driving energy cycles. The estimation of filtered probabilities through the MS-BVAR model provides valuable insights into the timing and duration of spe-

cific energy cycles used in this study, which the following equations can represent:

$$Y_{RE,t} = \alpha_a(s_t) + \sum_{p=1}^P \beta_{a,p}(s_t) Y_{RE,t-p} + \varepsilon_{RE,t}(s_t) \quad (1)$$

$$Y_{FS,t} = \alpha_b(s_t) + \sum_{p=1}^P \beta_{b,p}(s_t) Y_{FS,t-p} + \varepsilon_{FS,t}(s_t), \quad (2)$$

where $Y_{RE,t} = (ECO_t, ERIX_t, SPGC_t)$ and $Y_{FS,t} = (BO_t, NG_t, CP_t)$ represent the vector of endogenous variables for the renewable energy sector and fossil energy sector, respectively, at time t . $\alpha(s_t)$ is the regime-dependent intercepts, $\beta(s_t)$ is the regime-dependent autoregressive coefficients of lagged endogenous variables. $\varepsilon_t(s_t) \sim N(0, \sum(s_t))$ is the regime-dependent error terms which are assumed to follow a normal distribution with mean zero and regime-dependent variance $\sum(s_t)$. s_t represents the two regimes or states of the market: the low volatility regime (market downturn) and the high volatility regime (market upturn). This binary regime variable ($s_t = 1, 2$) indicates the state of the market at each point in time and is used to model the switching behavior of the system. However, s_t is unobservable, and the probability of being in each regime is regulated by a first-order Markov chain, which means that the probability of transitioning from one regime to another depends only on the current regime and not on past regimes. The transition probability between two regimes in the MS-BVAR model can be represented by.

$$p_{lh} = Pr(S_t = l | S_t = h) \text{ and } \sum_{l=1}^2 p_{lh} = 1, \quad l, h = 1, 2, \quad (3)$$

where p_{lh} is the probability of switching from regime l to regime h . Therefore, by defining two regimes, a transition probability can be computed in the form of a transition matrix as the following equation:

$$Q = \begin{bmatrix} p_{11} & 1 - p_{22} \\ 1 - p_{11} & p_{22} \end{bmatrix}. \quad (4)$$

Using the transition probabilities, we can calculate the expected duration of regimes 1 and 2 as follows

$$D(s_t = 1) = \frac{1}{1 - p_{11}}, D(s_t = 2) = \frac{1}{1 - p_{22}}. \quad (5)$$

3.2.2. Bayesian estimation

In this study, Bayesian estimation is used to estimate all unknown parameters in the MS-BVAR model. Let $\Theta(s_t) = (\alpha(s_t), \beta(s_t), \sum(s_t))$. The complete posterior distribution is constructed as the product of the likelihood function, and the prior distribution as

$$P(\Theta(s_t), Q | Y_t) \propto P(Y_t | \Theta(s_t), Q) P(\Theta(s_t), Q), \quad (6)$$

where $P(Y_t | \Theta(s_t), Q)$ is the product of the regime-dependent conditional normal likelihood function and filtered probabilities (Kim and Nelson, 1999).

Table 3
Result of unit root test.

Variables	Levels		First differences	
	ADF test	PP test	ADF test	PP test
lnECO	-1.5985	-1.7758	-11.1711***	-11.3013***
lnERIX	-3.005	-3.0102	-12.1867***	-12.2838***
lnSPGC	-1.8457	-1.9397	-11.1369***	-11.1551***
lnBO	-2.0111	-1.6954	-10.0868***	-9.9308***
lnNG	-1.8250	-2.7021	-16.2240***	-16.8181***
lnCP	-0.7951	-1.4242	-10.2897***	-10.4223***
lnREPU	0.9594	0.3353	-11.6332***	-39.0017***
lnUSD	-3.0827	-2.3935	-4.0793***	-12.9071***
lnINR	-2.0148	-1.9728	-10.4904***	-10.5256***
lnIPI	-2.9249	-3.0296	-10.5533***	-10.3502***

Note: *** denotes rejecting the null hypothesis at 1%.

$$P(Y_t | \Theta(s_t), Q) = \prod_{t=1}^T \left(\sum_{s_t} P(Y_t | Y_{t-1}, \Theta(s_t)) P(s_t = l | Y_{t-1}, \Theta(s_t), Q) \right) \quad (7)$$

$P(s_t = l | Y_{t-1}, \Theta(s_t), Q)$, for $t = 1, \dots, T$ and $l = 1, 2$, represents the filtered probabilities that are unknown and need to be estimated in the MS-BVAR model. These filtered probabilities provide information about the likelihood of being in each regime at a given time, based on the observed process for variable Y_t . The estimation of $P(s_t = l | Y_{t-1}, \Theta(s_t), Q)$ is carried out using a forward-filtering and backward-smoothing procedure, following the approach discussed by Kim and Nelson (1999). For the prior distribution of regime-dependent parameters $P(\Theta(s_t), Q)$, the independence of the prior distribution of the state-dependent parameters and transition matrix is assumed. We assume the following prior specification. The Sims-Zha prior (Sims and Zha, 1998) is assumed for $\beta(s_t)$, The Dirichlet prior is assumed for Q , and the inverse-Wishart prior is assumed for $\sum(s_t)$.

To estimate the joint posterior distribution of all parameters in the model, this study employs the Gibbs sampler due to the non-standard form of the posterior distribution (Chib, 1996). The Gibbs sampler performs 100,000 rounds of parameter sampling, discarding the initial 20,000 rounds as they represent nonstationary samples. The remaining 80,000 cycles are used to compute a specific value, which is determined as the mean of the 80,000 calculated values. This approach allows for robust estimation of the model parameters and provides reliable inference for the analysis.

Bayesian estimation surpasses conventional estimation in both accuracy and reliability. Incorporating a prior distribution or the researcher's expert opinion into the likelihood function enables the estimation of complex models, including nonlinear models like MS-VAR, that may otherwise be difficult to estimate. In particular, Bayesian estimation offers a solution to the issue of over-parameterization commonly encountered in MS-VAR models, where parameter estimates can be excessively high.

Bayesian approximations are particularly well-suited to small

datasets, relying on a sample-based method to estimate model parameters. In contrast, Maximum Likelihood estimation typically requires more significant amounts of data and an optimization method. The MS-BVAR model has been widely applied to the study of cycles in economics, such as in Krolzig (2013), as well as in the energy price, as evidenced by recent studies from Gong et al. (2021) and Shahrestani and Rafei (2020). This study aims to estimate renewable and fossil energy price cycles using filtered probabilities derived from the filtering process in the MS-BVAR model.

3.2.3. Multiple regression model

To examine the influence of the conflict between Ukraine and Russia on renewable and fossil energy price cycles. This study employs Multiple regression to robustness check. In this model, consider renewable and fossil energy prices cycle in regime 2 (High volatility) that computed from MS-BVAR as a dependence variable which can be written as the following equation:

$$REC_t = \beta_0 + \beta_1 \ln REPU_t + \beta_2 Dum_t + \beta_3 \ln REPU_t \times Dum_t + \beta_4 \ln USD_t + \beta_5 \ln INR_t + \beta_6 \ln IPI_t + \varepsilon_t \quad (8)$$

$$FEC_t = \theta_0 + \theta_1 \ln REPU_t + \theta_2 Dum_t + \theta_3 \ln REPU_t \times Dum_t + \theta_4 \ln USD_t + \theta_5 \ln INR_t + \theta_6 \ln IPI_t + \varepsilon_t, \quad (9)$$

where REC_t and FEC_t are filtered probabilities in regime 2 (High Volatility) of renewable and fossil energy cycles, respectively, obtained from the MS-BVAR. This model includes $\ln REPU_t \times Dum_t$, interaction terms to identify differential Ukraine-Russia conflict effects on renewable and fossil energy prices.

4. Empirical results and discussion

The following section presents the results of the analysis, which are divided into five parts: (1) unit root test, (2) model selection, (3) estimation results from the MS-BVAR model, (4) regression estimation results, and (5) impulse response analysis.

4.1. Unit root and cointegration tests

The first part of the analysis involves conducting unit root tests, specifically the Augmented Dickey-Fuller (ADF) test and Philips-Perron (PP) test, to assess the stationarity of the logarithmic data. This step is crucial before applying the Markov-Switching Bayesian Vector Autoregressive (MS-BVAR) and multiple regression models. The unit root test results are summarized in Table 3.

Table 3 presents the results of the unit root tests performed using the Augmented Dickey-Fuller (ADF) and Philips-Perron (PP) tests. The findings indicate that all logarithmic variables were nonstationary at the level or I(0). However, the null hypothesis was rejected, and the alternative hypothesis was accepted after applying a first-order difference or I (1) transformation. Consequently, all variables in this study were

Table 4
Johansen's cointegration test.

Null	Alternative	Eigenvalue	p-value	Trace	p-value
Renewable energy					
$r = 0$	$r \leq 1$	121.0247	0.0000	125.6383	0.0000
$r = 1$	$r \leq 2$	4.5442	0.7979	4.6135	0.8484
Fossil energy					
$r = 0$	$r \leq 1$	24.9707	0.0000	38.9811	0.0000
$r = 1$	$r \leq 2$	10.9619	0.1561	16.0104	0.1087

Note: Null and Alternative denote null and alternative hypotheses, respectively. r is a number of cointegrations.

Table 5
Model selection criteria.

Model	Renewable energy	Fossil energy
DIC		
BVAR (1)	−7.5644	−5.7083
BVAR (2)	−6.7914	−5.1830
MS-BVAR (1)	−7.6503	−6.4844
MS-BVAR (2)	−7.2324	−6.0924

Note: The bold number is the lowest value. The number inside the parentheses indicates the lag order.

deemed stationary at I (1) with a significance level of 0.01, indicating that they are nonstationary in levels but stationary after first differencing.

To assess the presence of long-run relationships among the variables, we conduct multivariate cointegration tests for both renewable and fossil energies. The results from the Johansen cointegration tests, using both the maximal eigenvalue and trace methods, are presented in Table 4. These tests confirmed the existence of significant long-run relationships among the variables in both energy groups, supporting the notion of cointegration. Notably, there exists a single cointegration vector for both energy groups. It is important to note that this study focuses on confirming the long-run relationships and validating the effectiveness of the MS-BVAR model in accurately capturing and filtering energy cycles. The analysis of short-run dynamics falls outside the scope of this research.

4.2. Model selection

Prior to estimating the data using the MS-BVAR model, it is essential to determine the appropriate lag and number of regimes, as these choices significantly impact model accuracy and reliability. In this study, we employ the Deviance Information Criterion (DIC), a widely recognized and effective metric for model evaluation in Bayesian analysis, to assess the suitability of different model specifications. By leveraging the information derived from the posterior samples obtained through our analysis, the DIC provides a robust measure of model performance and aids in selecting the most appropriate specification (François and Laval, 2011). Table 5 compares the linear BVAR and the

nonlinear MS-BVAR models based on the DIC. Notably, the DIC analysis reveals that the nonlinear MS-BVAR (1) model is the superior choice for both renewable and fossil energy. This finding underscores the significance of incorporating nonlinear dynamics when capturing the intricate patterns and complexities inherent in the energy data under investigation.

4.3. Estimation and analysis of energy cycles using the MS-BVAR(1) model

The estimated results of the MS-BVAR (1) are presented in Table 6 as follows.

Table 6 presents compelling results obtained from the MS-BVAR (1) model for renewable energy prices, specifically ECO, ERIX, and SPGC, as well as fossil energy prices, namely BO, NG, and CP. These results provide robust evidence supporting the presence of distinct regimes in the energy market and shed light on their associated dynamics. Notably, the intercept values for Regime 2 ($\alpha_a(s_t = 2)$ and $\alpha_b(s_t = 2)$) consistently exhibit a dominant positive trend, surpassing the intercept values for Regime 1 ($\alpha_a(s_t = 1)$ and $\alpha_b(s_t = 1)$). This clear disparity underscores the substantial divergence in energy price patterns and market behavior between the two regimes. The findings strongly suggest that Regime 1 corresponds to a high energy price or high volatility regime. Within this regime, energy prices experience significant fluctuations, indicative of a market environment characterized by heightened uncertainty, rapid changes, and potential price shocks. On the other hand, Regime 2 represents a low energy price or low volatility regime characterized by relatively stable and predictable market conditions (Gong et al., 2021).

When examining the probabilities of remaining in each energy price regime, the parameters p_{11} and p_{22} provide crucial insights into the persistence of low-volatility and high-volatility regimes, respectively. The analysis reveals compelling findings for both renewable and fossil energy prices. For renewable energy prices, the estimated value of p_{11} is 0.9906, indicating a remarkable 99.06% probability of remaining in a low volatility regime. This suggests that the price cycle of renewable energy is highly likely to exhibit stability and remain in a regime characterized by relatively predictable and less volatile price movements. In contrast, the estimated value of p_{22} is 0.9909, signifying an overwhelming 99.09% probability of remaining in a high volatility

Table 6
Results of MS-BVAR (1) model.

Dependent variable	Renewable energy cycle (REC_t)					
	Regime 1			Regime 2		
	$\ln ECO$	$\ln ERIX$	$\ln SPGC$	$\ln ECO$	$\ln ERIX$	$\ln SPGC$
$\alpha_a(s_t)$	−1.6396*** [0.1122]	0.4410*** [0.0249]	0.6258*** [0.0140]	0.3531*** [0.0635]	1.1787*** [0.0440]	1.6801*** [0.0246]
$\ln ECO_{t-1}$	0.7795*** [0.0459]	−0.0005 [0.0615]	0.2675*** [0.0241]	1.0114*** [0.0260]	0.0810*** [0.0660]	0.3706*** [0.0259]
$\ln ERIX_{t-1}$	0.5609*** [0.0145]	1.0014*** [0.0251]	0.0973*** [0.0521]	0.0108 [0.0135]	0.7800*** [0.0270]	−0.0150 [0.1122]
$\ln SPGC_{t-1}$	−0.2154*** [0.0440]	−0.0674*** [0.0115]	0.6456*** [0.0213]	−0.0718*** [0.0204]	0.0163 [0.0246]	0.5407*** [0.0459]
p_{11}	0.9906					
p_{22}	0.9909					

Dependent variable	Fossil energy cycle (FEC_t)					
	Regime 1			Regime 2		
	$\ln BO$	$\ln NG$	$\ln CP$	$\ln BO$	$\ln NG$	$\ln CP$
$\alpha_b(s_t)$	0.1034** [0.0553]	−0.3235*** [0.0330]	0.2878*** [0.0119]	0.6230*** [0.1112]	6.1705*** [0.0663]	0.2280** [0.0239]
$\ln BO_{t-1}$	0.9088*** [0.0229]	0.1575*** [0.0770]	−0.0233** [0.0459]	0.5218*** [0.0461]	0.4987** [0.3846]	−0.0425 [0.2292]
$\ln NG_{t-1}$	−0.0315*** [0.0166]	0.7138*** [0.0319]	0.0523*** [0.0612]	0.0197 [0.0827]	0.5005*** [0.1593]	0.0368 [0.1112]
$\ln CP_{t-1}$	0.0655*** [0.0365]	0.0239 [0.0132]	0.9942*** [0.0253]	0.4598*** [0.0663]	1.0331*** [0.0239]	0.9688*** [0.0461]
p_{11}	0.9670					
p_{22}	0.9166					

Note: [] indicates the Standard Error, ***, **, and * denote significance levels at 0.01, 0.05, and 0.10, respectively. To confirm the robustness of our results, we have included Appendix Table A1, which demonstrates that our MS-BVAR findings remain consistent even when we consider NBP as an indicator for natural gas. This further strengthens the reliability of our results and underscores the stability of our analysis.

regime. This suggests that renewable energy prices are prone to experiencing significant fluctuations and unpredictable price movements within a high-volatility regime. Moreover, the analysis reveals a probability of transitioning from a low volatility to a high volatility regime, as indicated by the value of p_{12} ($1 - 0.9906 = 0.0094$). This implies a relatively small but non-negligible 0.94% chance of renewable energy prices transitioning from a stable regime to a more volatile regime. Similarly, the estimated value of p_{21} ($1 - 0.9909 = 0.0091$) suggests an extremely low 0.91% chance of transitioning from a high volatility to a low volatility regime.

Turning to fossil energy prices, the estimated values of p_{11} and p_{22} are 0.9220 and 0.8558, respectively. These values indicate a higher probability of remaining in a low volatility regime (92.20%) and a high volatility regime (85.58%) for fossil energy prices. This suggests that fossil energy prices exhibit a relatively greater degree of stability, with lower volatility than renewable energy prices. Similarly, the analysis reveals probabilities of transitioning between volatility regimes for fossil energy prices. The value of p_{12} ($1 - 0.9220 = 0.0780$) signifies a 7.80% chance of transitioning from a low volatility to a high volatility regime. Conversely, the estimated value of p_{21} ($1 - 0.8558 = 0.1442$) suggests a 14.42% chance of transitioning from a high volatility to a low volatility regime. These findings highlight significant differences in the volatility dynamics of renewable and fossil energy prices. Renewable energy prices show a higher persistence in either a low or high volatility regime, indicating greater stability or fluctuation. On the other hand, fossil energy prices display less persistence, with a lower tendency to remain in the same volatility regime over time. This suggests that renewable energy prices are more likely to maintain their current level of volatility, while fossil energy prices are subject to more frequent changes in volatility (Paiva et al., 2018). The higher volatility observed in fossil energy prices compared to renewable energy prices can be attributed to the exhaustible and finite nature of fossil fuel sources, such as oil and natural gas. As these resources become increasingly scarce, their extraction and production become more challenging and costly (Schumacher, 1985). The limited availability of fossil fuels, coupled with fluctuations in market conditions and supply, creates higher price volatility in the fossil energy market (Shahrestani and Rafei, 2020). In contrast, renewable energy sources, characterized by their abundance and sustainability, offer a more stable and predictable market environment (Khan and Su, 2022). In contrast, renewable energy sources, such as solar, wind, and hydroelectric power, are generally more abundant and sustainable. The availability of these resources is not subject to the same geopolitical risks or supply constraints as fossil fuels (Paiva et al., 2018). Additionally, advancements in renewable energy technologies and infrastructure have led to more predictable and stable production,

reducing the inherent volatility in renewable energy prices.

To deepen our understanding of price cycles in fossil and renewable energy markets, we employ Kim and Nelson's forward-filtering and backward-smoothing procedures Kim and Nelson (1999) within the MS-BVAR model. This approach allows us to examine the filtered probabilities of staying in Regime 2, characterized by high volatility or a high-price regime. Figs. 1 and 2 present a visual representation of these probabilities. Fig. 1 focuses on the energy price cycle of fossil fuels, while Fig. 2 highlights the energy price cycle of renewable energy sources. The left vertical axis denotes probability values ranging from 0 to 1, providing valuable insights into the likelihood of being in the high volatility regime. The horizontal axis encompasses the period from January 2010 to December 2022, enabling a comprehensive understanding of the price dynamics over time. Additionally, including the Russia Economic Policy Uncertainty Index on the right axis offers a contextual reference.

Moreover, to assess the robustness of our findings, we incorporate an additional element into the analysis by including a dummy variable representing the conflict between Ukraine and Russia. This dummy variable is constructed based on Google Trends data, measuring the search popularity of the phrase "Ukraine and Russia Tensions." By plotting the energy cycles alongside the dummy variable and the Russia Economic Policy Uncertainty (EPU) index, we aim to examine the synchronization and potential influence of geopolitical events on the energy market. This robustness check allows us to evaluate the consistency of the observed patterns and further explore the relationship between energy prices and geopolitical uncertainties. The Figure includes the fossil energy cycle (indicated by a dark brown line), the renewable energy cycle (indicated by a blue line), the Russia Economic Policy Uncertainty index (indicated by a red line), and the dummy variable (indicated by an orange line and area).

Fig. 1 provides a visual representation of the price cycle of fossil energy, offering supporting evidence for the hypothesis that the cyclical nature of fossil energy prices is influenced by the Ukraine-Russia conflict at specific time points. Notably, there is a discernible increase in the fossil energy cycle observed during the period of 2014–2017, which aligns with significant events such as the annexation of Crimea in March 2014, the downing of Malaysian Airlines flight MH17 in July 2014, and the ongoing war in Eastern Ukraine until 2017. The conflict in Eastern Ukraine ultimately resulted in a ceasefire by the end of 2019 following top-level talks (Bowen, 2019).

During the conflict between Ukraine and Russia, the cutoff of natural gas supplies from Russia to Ukraine raised concerns about potential energy crises in Europe, given the heavy reliance of the European Union on Russian natural gas imports. The disruption in gas supplies created

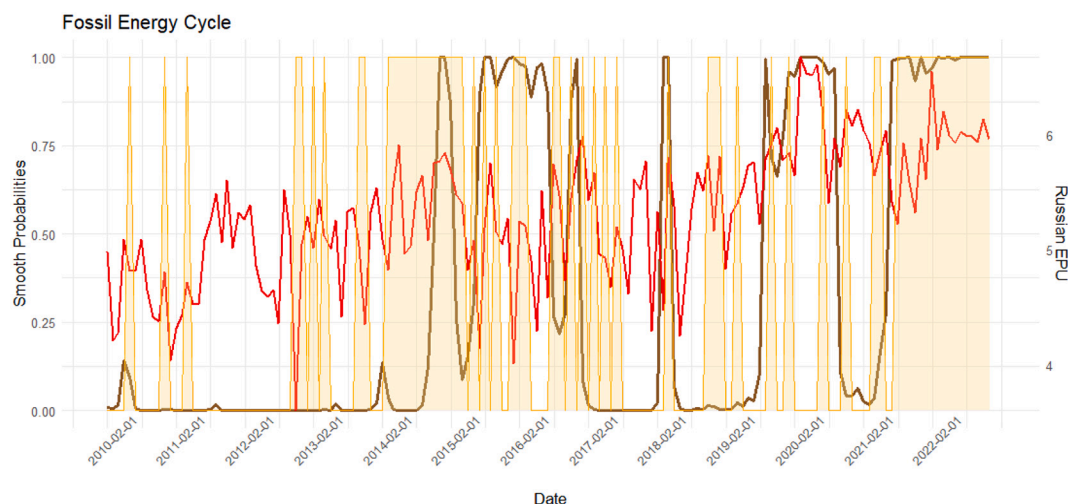


Fig. 1. Fossil energy cycle from the period 2010/1 to 2022-12.

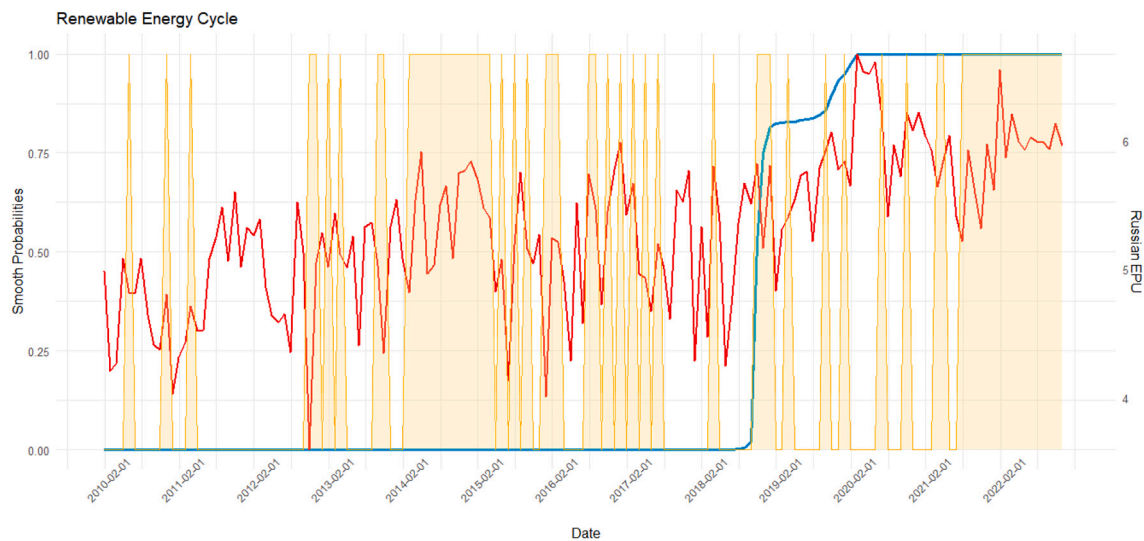


Fig. 2. Renewable energy cycle from the period 2010/1 to 2022–12.

uncertainties in the energy market, impacting natural gas prices and availability, which had broader effects on energy prices (Natorski and Pomorska, 2017). Another significant event was the Russian invasion of Ukraine in February 2022. This event profoundly impacted the fossil energy price cycle, with a dramatic increase observed during this period. The price cycle exhibited strong persistence in staying in the high volatility regime, reflecting the disruptive influence of geopolitical tensions. These disruptions encompassed the loss of crucial resources such as crude oil, natural gas, and coal fields and interruptions in production and transportation. As a result, energy shortages, higher prices, and increased volatility were observed in the fossil energy markets (Umar et al., 2022). Notably, this period aligns with the dummy variable representing the Ukraine-Russia conflict, further supporting the connection between geopolitical events and energy market dynamics.

Fig. 2 presents the cycle of high volatility in renewable energy prices. From 2010 until mid-2018, renewable energy prices exhibited relative stability with low volatility, seemingly unaffected by the conflict between Russia and Ukraine. However, a noticeable renewable energy price cycle shift towards a high volatility regime emerged in late 2018.

This shift coincided with many countries' increased adoption of renewable energy, indicating that factors other than the Ukraine-Russia conflict may have influenced renewable energy prices during this period. It is worth noting that the persistence observed in the renewable energy cycle in 2022 may be attributed to the REpowerEU policy. This policy was drafted to reduce dependence on Russian energy and ensure future energy security within the European Union. Implementing such policies reflects efforts to promote renewable energy and address energy security concerns within the context of geopolitical tensions.

This analysis further supports the existing evidence that conflicts between Ukraine and Russia exert a more pronounced influence on fossil energy prices than renewable energy prices. This finding aligns with previous studies by Regnier (2007) and Dinçer et al. (2018) that have investigated the volatility of the fossil energy market. The ability of fossil energy producers to adjust production levels gives them significant leverage over the volatility of fossil energy prices. Consequently, conflicts that disrupt fossil energy supply, such as those involving Russia, can lead to substantial price fluctuations. On the other hand, renewable energy prices demonstrate a higher degree of resilience and are less

Table 7

The impact of EPU on energy price cycles.

Fossil energy cycle	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Variables	Coef.	Coef.	Coef.	Coef.	Coef.	Coef.
Intercept	0.2366*** [0.0445]	−1.2610*** [0.2931]	−0.8505** [0.3602]	−8.1517** [3.3678]	−7.6318** [3.3967]	−3.6639** [1.4560]
lnREPU		0.3012*** [0.0547]	0.2089*** [0.0687]		0.0408*** [0.0173]	0.0699*** [0.0184]
Dum	0.2546*** [0.0693]		−0.6811 [0.4221]	0.0608* [0.0296]		−0.9373 [0.5328]
lnREPU x Dum			0.1595*** [0.0405]			0.2082*** [0.0992]
lnUSD				2.7155*** [0.3243]	2.7171*** [0.3556]	2.6197*** [0.3443]
lnINR				−0.2200*** [0.0728]	−0.1897** [0.0814]	−0.2620*** [0.0823]
lnIPI				1.5481 [0.9525]	0.9428 [0.9553]	1.5893 [0.9502]

Renewable energy cycle	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Variables	Coef.	Coef.	Coef.	Coef.	Coef.	Coef.
Intercept	0.2380*** [0.0462]	−2.2455*** [0.2494]	−2.0805*** [0.3148]	−1.9531*** [0.4375]	−7.2722** [3.1960]	−6.0485* [3.3633]
lnREPU		0.4794*** [0.0465]	0.4456*** [0.0601]		0.3008*** [0.0536]	0.2321*** [0.0666]
Dum	0.1639 [0.0920]		−0.3970 [0.5436]	0.1044 [0.0582]		−0.8154 [0.5184]
lnREPU x Dum			0.0789 [0.1003]			0.1587 [0.0965]
lnUSD				1.9383*** [0.3433]	1.3139*** [0.3346]	1.2574*** [0.3551]
lnINR				−0.4338*** [0.0771]	−0.2626*** [0.0776]	−0.3046*** [0.0800]
lnIPI				0.4127 [0.9025]	0.0555 [0.8048]	0.0741 [0.8272]

Note: [] indicates the Standard Error, ***, **, and * denote significance levels at 0.01, 0.05, and 0.10, respectively. Dummy denotes the time point of conflict between Ukraine-Russia.

susceptible to immediate impacts resulting from such conflicts. In addition, recent studies conducted by Czech and Wielechowski (2021) and Hemrit and Benlagha (2021) emphasized the interconnected nature of fossil and renewable energy markets. These studies highlight the existence of substitute products and investment strategies to mitigate risks associated with fluctuations in fossil energy prices. The interdependence between these two energy sectors underscores the significance of renewable energy as a potential safeguard against volatility in the fossil energy market (Karkowska and Urjasz, 2023a).

To strengthen the understanding of the relationship between the Ukraine-Russia conflict and energy price cycles, this study conducts additional analysis using a regression model described in Section 4.4. This regression model is employed to examine the nexus between the conflict and the observed patterns in energy price cycles. By quantitatively assessing the relationship, the study aims to provide empirical evidence supporting the influence of the Ukraine-Russia conflict on energy prices. The results of this regression analysis are presented and discussed in the following section, shedding further light on the dynamics between geopolitical events and energy market fluctuations.

4.4. Empirical results of regression models

Table 7 presents the regression estimation results for assessing the relationship between the Ukraine-Russia conflict and energy price cycles, specifically focusing on the fossil and renewable energy cycles. In Columns (1) and (2), the estimates are reported with the inclusion of the dummy and EPU variables, respectively, representing the Ukraine-Russia conflict. To capture potential interactions, Column (3) includes the interactions between the dummy variable and Russia's Economic Policy Uncertainty Index (REPU). Controlling for omitted-variable bias, Columns (4), (5), and (6) incorporate additional control variables, namely the US Dollar Index, Interest rate, and Industrial Production Index. The upper panel of the table presents the results for the renewable energy cycle, while the lower panel displays the results for the fossil energy cycle.

Based on the regression analysis conducted, our findings consistently demonstrate a stronger and statistically significant impact of the Ukraine-Russia conflict on fossil energy price cycles compared to renewable energy price cycles. Both model specifications (columns 1 and 2) provide robust evidence supporting this conclusion, as indicated by the positive and significant coefficient of 0.2456 for the conflict dummy variable and 0.3012 for the REPU variable in relation to fossil energy cycles. This result highlights the substantial influence of the conflict on the dynamics of the fossil energy price cycle.

Furthermore, when examining the interaction between the conflict dummy variable and REPU, we observe that the fossil energy price cycle coefficient remains statistically significant with a coefficient of 0.1595. This indicates that including the conflict dummy variable strengthens the positive influence of REPU on investments in the fossil energy price cycle. In simpler terms, REPU exerts a more substantial impact on the fossil energy price cycle during periods of the Ukraine-Russia conflict. The impact of economic uncertainty on energy prices during conflict periods is also heightened by approximately 0.1595. This suggests that economic uncertainty has a more pronounced effect on energy prices when conflict is ongoing. These results remain consistent when we include control variables in Column (6), with a coefficient of 0.2082. This consistent pattern of significance reaffirms our findings' reliability and emphasizes the conflict's influential role in shaping the fossil energy price cycle.

On the contrary, our regression results consistently indicate the absence of significant effects associated with the conflict dummy variable or its interaction with REPU on renewable energy cycles, as presented in Table 7. Our analysis reveals that REPU does indeed have a

positive influence on renewable energy prices. However, the interaction coefficient is statistically insignificant, suggesting that the positive impact of REPU on the renewable energy price cycle diminishes during the Russia-Ukraine conflict. The insignificant impact of the Ukraine-Russia war on renewable energy can be attributed to several plausible reasons. Firstly, the heavy dependence of many countries on imported fossil fuels, particularly natural gas and oil, from Russia and Ukraine renders them more vulnerable to supply chain disruptions stemming from the conflict. Consequently, these disruptions can lead to reduced availability and increased prices of fossil fuels (Zhou et al., 2023). In contrast, the renewable energy sector encompasses a diverse array of sources, including solar, wind, hydro, and geothermal power. Unlike fossil fuels, primarily traded as commodities, renewable energy sources exhibit greater geographical distribution and variety (Mohammed et al., 2023). Secondly, comprehensive government policies and incentives that foster stability and facilitate long-term investments underpin the renewable energy market. These policy frameworks provide a protective buffer, allowing the sector to withstand better short-term shocks associated with events such as the Ukraine-Russia conflict. Thirdly, the renewable energy market exhibits a degree of inertia resulting from intricate long-term planning, substantial infrastructure investments, and the involvement of multiple stakeholders (Steffen and Patt, 2022). This inherent inertia constrains the pace and magnitude of policy changes, even in the face of external shocks like the Ukraine-Russia war.

The inclusion of control variables in the analysis provides valuable insights into the factors influencing both renewable and fossil energy cycles. The findings reveal that the US Dollar index significantly impacts both renewable and fossil energy cycles, indicating its role as a driver of energy market dynamics. The US Dollar index serves as a measure of the strength of the US dollar relative to other major currencies, and its influence on energy markets can be attributed to its impact on global trade, investments, and commodity pricing.

4.5. Estimation results of the energy prices response to the Ukraine-Russia conflict

Finally, in this section, we analyze the generalized impulse response function extended from the MSBVAR model with additional Ukraine-Russia conflict ((lnREPU x Dum) variable to investigate the impact of the Ukraine-Russia conflict ((lnREPU x Dum) on fossil and renewable energy prices in low and high volatility regimes. This analysis is a valuable tool for studying the dynamic relationship between variables and capturing the short-term effects of shocks. In a conflict, the impulse response function can help identify how renewable and fossil energy prices respond to an exogenous shock, such as the Ukraine-Russia conflict. It allows us to trace the immediate impact and subsequent adjustments in the energy market following the shock. The results of this analysis are visually presented in Figs. 3 and 4, providing insights into the dynamics of energy prices in response to the Ukraine-Russia conflict.

Fig. 3 displays the response of renewable energy prices to the shock induced by the Ukraine-Russia conflict, categorized into high and low-volatility regimes. The left panel illustrates the shock response in low-volatility regimes, while the right panel depicts the response in high-volatility regimes. The findings depicted in Fig. 3 reveal that the response of renewable energy prices is characterized by a relatively muted effect in both volatility regimes. In the low volatility regime, the response of renewable prices shows a minor and negative reaction during the first three months following the shock. This negative response suggests a temporary decline in renewable energy prices in the early stages of the conflict. However, the response gradually converges to equilibrium as time progresses, indicating a return to pre-shock conditions. In contrast, the high volatility regime exhibits a slightly positive initial response to renewable energy prices. This positive reaction

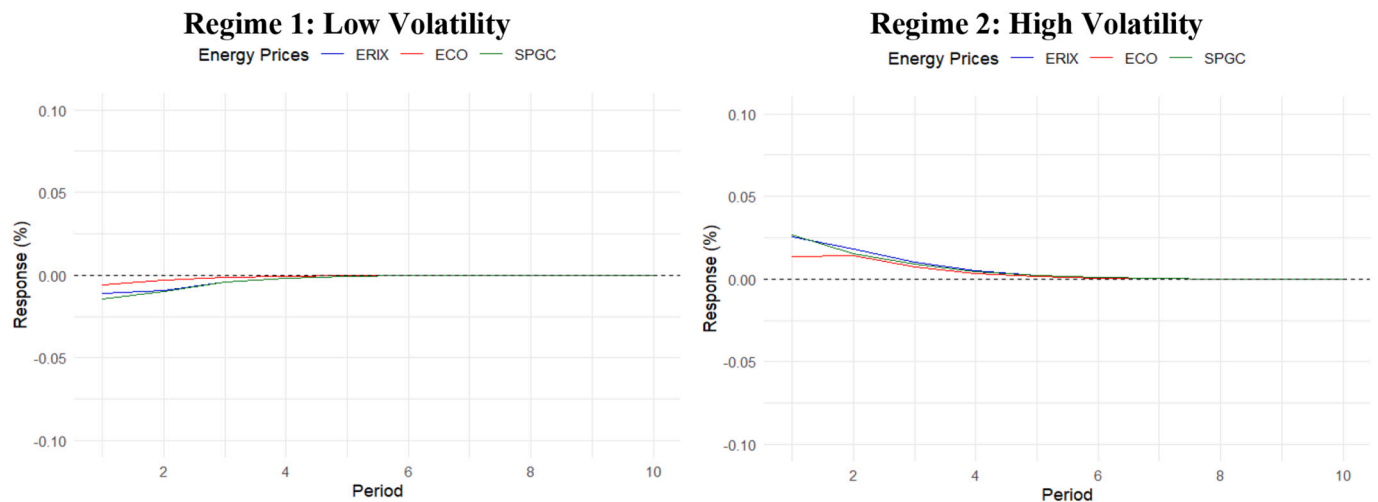


Fig. 3. Generalized impulse response of renewable energy prices to the Ukraine-Russia conflict.

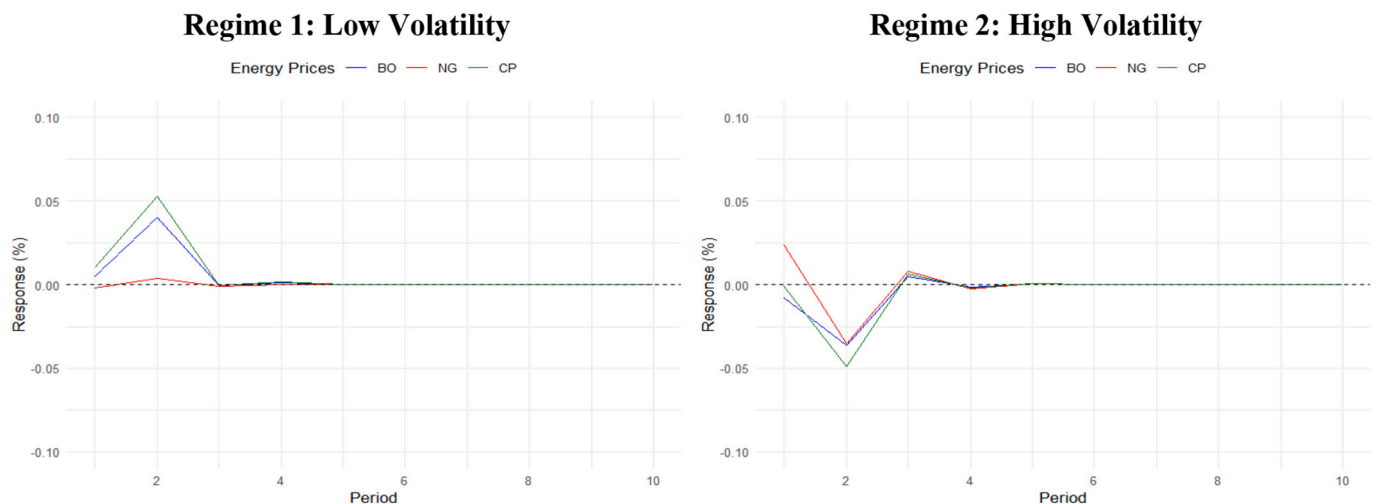


Fig. 4. Generalized impulse response of fossil energy prices to the Ukraine-Russia conflict.

indicates a modest increase in renewable prices immediately after the shock in high volatility periods. However, similar to the low volatility regime, the response ultimately converges to the equilibrium, signifying a stabilization of renewable energy prices over the long run.

Fig. 4 displays the response of fossil energy prices to the shock stemming from the Ukraine-Russia conflict. The results indicate a consistent pattern across all fossil price responses, whereby the coal price (CP) exhibits the highest response in the low volatility regime. The responses of these fossil prices reach their peak in the second month following the shock before gradually adjusting towards equilibrium by the sixth month. This pattern suggests that the shock of the Ukraine-Russia conflict initially led to an increase in fossil energy prices, with the impact being most pronounced in the early stages. However, as time progresses, the prices gradually converge towards their pre-shock levels, indicating a return to equilibrium conditions. In contrast, the response of fossil prices in the high volatility regime shows a negative trend in the aftermath of the Ukraine-Russia conflict shock. The negative response reaches its highest point in the second month before eventually converging to equilibrium by the sixth month. This implies that, in high volatility periods, the shock initially leads to a decrease in fossil energy prices, followed by a gradual return to pre-shock levels.

We have performed robustness analyses and have visualized the impulse responses generated through the Local Projections Impulse

Response Functions, employing the MS-BVAR model, to facilitate comparison. These results have been presented in Figs. A1 and A2. Our findings illustrate that these additional results do not significantly alter the relationship between the REPU shock and energy prices. These outcomes further bolster the confidence in the stability and reliability of our impulse response function.

5. Conclusion and policy recommendation

This study represents a significant contribution to the field by pioneering an investigation into the influence of the Ukraine-Russia conflict on energy cycles, an area of research that has remained largely unexplored. By adopting an innovative approach that draws upon the concept of business cycles, traditionally focused on economic fluctuations, this study extends the analysis to the energy sector, shedding new light on the intricate dynamics of energy price cycles and their susceptibility to the Ukraine-Russia conflict. To achieve this research objective, a robust methodology is employed, utilizing the Markov Switching Bayesian Vector Autoregressive (MS-BVAR) model to filter the cycle of each energy market. Additionally, the Russia Economic Policy Uncertainty Index, alongside carefully constructed dummy variables derived from Google search data, is incorporated to investigate the impact of the Ukraine-Russia conflict on energy cycles using a regression approach.

This comprehensive framework incorporates a broader range of conflicts, energy system characteristics, and contextual conditions compared to previous frameworks, building upon and expanding existing knowledge in the field.

By employing robust methodologies, our study has uncovered compelling and significant insights, which we present as follows: First, our study has yielded intriguing and meaningful findings, which we present as Our examination of price cycles utilizing the MS-BVAR model has unveiled striking differences between renewable and fossil energy prices. Notably, renewable energy prices exhibit a remarkable degree of persistence, often residing within either low or high-volatility regimes for extended durations. Between 2010 and 2018, renewable energy prices predominantly lingered within the low volatility regime, indicative of a market downturn. However, from 2019 to 2022, we observed a shift towards heightened fluctuations, signaling a market upswing. Conversely, fossil energy prices display less persistence, demonstrating a lesser propensity to remain within the same volatility regime over time. Notably, we observed substantial regime shifts coinciding with the escalation of the Ukraine-Russia conflict. Our findings show renewable and fossil energy prices' distinct characteristics and response dynamics. Second, the results from our regression analysis provide empirical confirmation that conflicts involving Ukraine and Russia exert a more pronounced influence on fossil energy prices compared to renewable energy prices. This indicates that the Ukraine-Russia conflict wields a more substantial impact on the pricing dynamics of fossil fuels, potentially due to disruptions in supply chains, geopolitical tensions, and market sensitivities. In contrast, the influence on renewable energy prices appears relatively constrained, suggesting a certain level of disconnection between the conflict and the renewable energy market. These findings contribute to a deeper comprehension of how conflicts mold energy markets and emphasize the differing effects on various energy sources. Lastly, our exploration of generalized impulse response functions uncovers that the reaction of renewable energy prices to the shock of the Ukraine-Russia conflict is relatively modest and transient, regardless of the prevailing volatility regime. These findings imply that the conflict's impact on renewable energy prices is short-lived, with the market swiftly adapting and returning to pre-shock conditions over time. In contrast, fossil energy prices exhibit a more pronounced and enduring response to the conflict shock, featuring consistent patterns of price adjustments across all types of fossil fuels. Notably, in the low volatility regime, coal prices exhibit the most substantial response. These results further emphasize the contrasting impacts of the Ukraine-Russia conflict on renewable and fossil energy markets.

Drawing from the findings of our study, we propose a set of policy recommendations that can effectively address the challenges posed by this conflict: First, in high-importing fossil energy countries, prioritizing the diversification of energy sources and supply routes is paramount. Policymakers should actively explore alternative avenues for energy imports, fostering strategic partnerships and collaborations to reduce dependence on specific regions or sources. Diversifying the energy portfolio, particularly away from regions susceptible to conflicts like

Ukraine and Russia, can significantly bolster the stability and reliability of energy systems. Second, policymakers in high-importing fossil energy nations should make substantial investments in enhancing the resilience and interconnectedness of their energy infrastructure. This encompasses the development of advanced transmission and distribution networks, robust storage facilities, and the adoption of cutting-edge smart grid technologies. Strengthening the efficiency and flexibility of energy systems empowers countries to absorb shocks and ensure uninterrupted energy market operations during periods of geopolitical instability. Third, establishing comprehensive and robust mechanisms for monitoring fossil energy markets and assessing risks arising from geopolitical conflicts is imperative. Policymakers must design sophisticated risk assessment frameworks that encompass both short-term disruptions and long-term impacts on energy cycles. A proactive approach to monitoring market dynamics and identifying potential risks enables timely interventions and the implementation of effective contingency plans to minimize disruptions and maintain stability in the energy market. Fourth, fossil fuel exporting countries should strategically diversify their economies by investing in renewable energy industries and other sustainable sectors. Reducing dependence on fossil fuel exports mitigates economic risks associated with fluctuations in global energy markets. Additionally, fostering regional cooperation and collaboration can enhance energy security and stability. This may involve joint efforts in developing renewable energy infrastructure, promoting cross-border electricity trade, and engaging in collaborative research and development initiatives. Lastly, countries heavily reliant on renewable energy should continue to prioritize and expand their domestic renewable energy capacity. Given that renewable energy markets are less susceptible to the impacts of the Ukraine-Russia conflict, policymakers should focus on supporting the development of renewable energy projects, incentivizing private sector investments, and implementing favorable regulatory frameworks. This strategic emphasis further fortifies their renewable energy sector and contributes to sustainable, low-carbon energy transitions.

Further research should consider incorporating socioeconomic factors alongside conflict variables. This approach can enhance our understanding of the intricate relationship between economic, political, and social factors that shape energy cycles during conflict.

CRedit authorship contribution statement

Paravee Maneejuk: Conceptualization, Methodology, Software, Supervision, Writing – review & editing. **Nuttaphong Kaewtathip:** Data curation, Writing – original draft, Visualization, Investigation. **Woraphon Yamaka:** Conceptualization, Methodology, Software, Validation, Writing – review & editing.

Acknowledgements

The paper is supported by the Center of Excellence in Econometrics, Faculty of Economics, Chiang Mai University.

Appendix A

Table A1

Results of MS-BVAR (1) model with NBP natural gas variables.

Dependent variable	Fossil energy cycle (FEC_t)					
	Regime 1			Regime 2		
	$\ln BO$	$\ln NG$	$\ln CP$	$\ln BO$	$\ln NG$	$\ln CP$
$\delta(s_t)$	0.0027 [0.0549]	−0.3664*** [0.0326]	0.1174* [0.0131]	0.6232*** [0.1130]	−5.9500*** [0.0670]	0.0401 [0.0270]
$\ln BO_{t-1}$	0.8702*** [0.0224]	0.0762 [0.0820]	−0.0746** [0.0486]	0.5618*** [0.0461]	0.5671** [0.3797]	−0.0482 [0.2252]
$\ln NG_{t-1}$	−0.0268** [0.0196]	0.8056*** [0.0334]	0.0294** [0.0621]	0.0183 [0.0907]	0.3357*** [0.1547]	0.0174 [0.1130]
$\ln CP_{t-1}$	0.0781*** [0.0368]	0.0106 [0.0148]	1.0143*** [0.0253]	0.4276*** [0.0670]	1.1465*** [0.0270]	1.0184*** [0.0461]
P_{ll}	0.9601					
P_{hh}	0.8964					

Note: NG is NBP natural gas spot price.

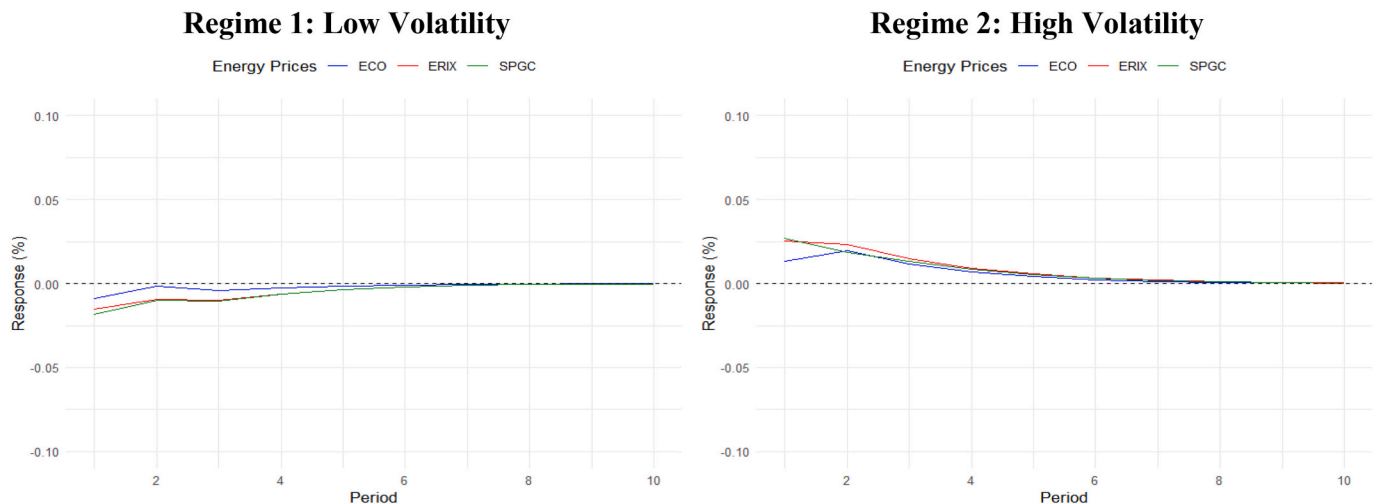


Fig. A1. Impulse response of renewable energy prices to the Ukraine-Russia conflict obtained from the local projections based on MS-BVAR.

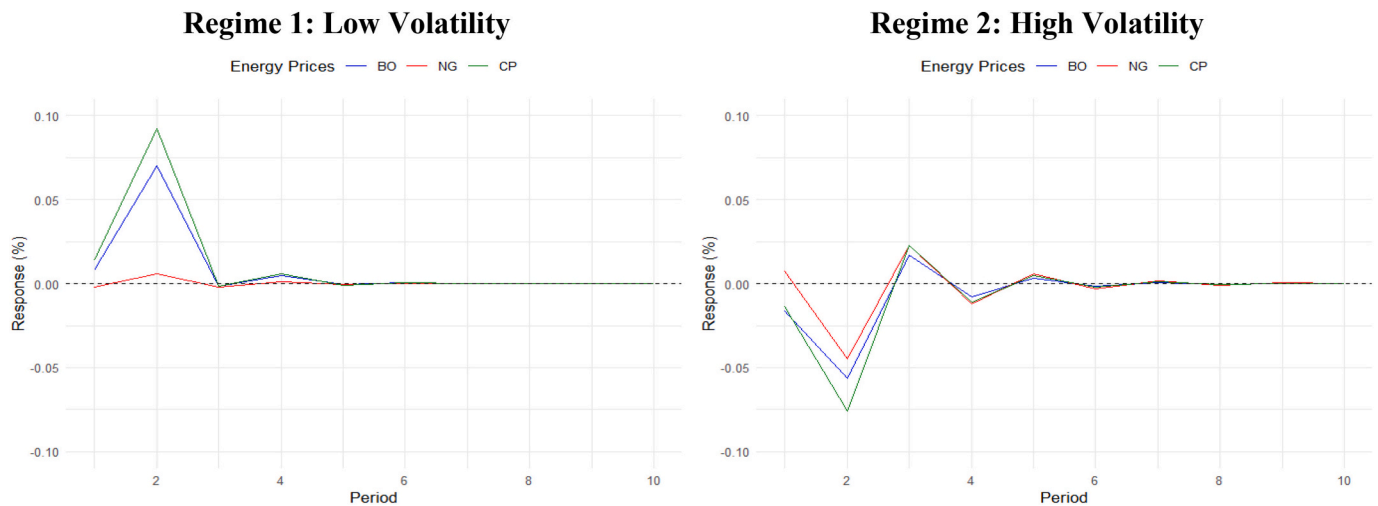


Fig. A2. Impulse response of fossil energy prices to the Ukraine-Russia conflict obtained from the local projections based on MS-BVAR.

Appendix B. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.eneco.2023.107218>.

References

- Agnello, L., Castro, V., Hammoudeh, S., Sousa, R.M., 2020. Global factors, uncertainty, weather conditions, and energy prices: on the drivers of the duration of commodity price cycle phases. *Energy Econ.* 90, 104862.
- Antonakakis, N., Chatziantoniou, I., Filis, G., 2014. Dynamic spillovers of oil price shocks and economic policy uncertainty. *Energy Econ.* 44, 433–447.
- Asl, M.G., Canarella, G., Miller, S.M., 2021. Dynamic asymmetric optimal portfolio allocation between energy stocks and energy commodities: evidence from clean energy and oil and gas companies. *Res. Policy* 71, 101982.
- Baker, S.R., Bloom, N., Davis, S.J., 2016. Measuring economic policy uncertainty. *Q. J. Econ.* 131 (4), 1593–1636.
- Benton, T.G., Froggatt, A., Wellesley, L., Grafham, O., King, R., Morissetti, N., Schröder, P., 2022. The Ukraine war and threats to food and energy security. In: Chatham House—International Affairs Think Tank, 2022-04.
- Berndt, E.R., Wood, D.O., 1975. Technology, prices, and the derived demand for energy. *Rev. Econ. Stat.* 259–268.
- Bowen, A.S., 2019. Coercive diplomacy and the Donbas: explaining Russian strategy in eastern Ukraine. *J. Strateg. Stud.* 42 (3–4), 312–343.
- Bulavskaya, T., Reynès, F., 2018. Job creation and economic impact of renewable energy in the Netherlands. *Renew. Energy* 119, 528–538.
- Burns, A.F., Mitchell, W.C., 1946. The basic measures of cyclical behavior. In: *Measuring Business Cycles*. NBER, pp. 115–202.
- Cashin, P., McDermott, C.J., 2002. The long-run behavior of commodity prices: small trends and big variability. *IMF Staff Papers* 49 (2), 175–199.
- Cheikh, N.B., Zaied, Y.B., 2023. Renewable energy deployment and geopolitical conflicts. *J. Environ. Manag.* 344, 118561.
- Chen, J., Ma, F., Qiu, X., Li, T., 2023. The role of categorical EPU indices in predicting stock-market returns. *Int. Rev. Econ. Financ.* 87, 365–378.
- Chib, S., 1996. Calculating posterior distributions and modal estimates in Markov mixture models. *J. Econ.* 75 (1), 79–97.
- Czech, K., Wielechowski, M., 2021. Is the alternative energy sector COVID-19 resistant? Comparison with the conventional energy sector: Markov-switching model analysis of stock market indices of energy companies. *Energies* 14 (4), 988.
- Dash, S.R., Maitra, D., 2021. Do oil and gas prices influence economic policy uncertainty differently: multi-country evidence using time-frequency approach. *Q. Rev. Econ. Financ.* 81, 397–420.
- de la Horra, L.P., Perote, J., de la Fuente, G., 2022. The impact of economic policy uncertainty and monetary policy on R&D investment: an option pricing approach. *Econ. Lett.* 214, 110413.
- De Villa, M.A., 2023. Assessing geopolitical risk: a multi-level approach for top managers of multinationals. *AIB Insights* 23 (1), 1–7.
- Deaton, A., Laroque, G., 1992. On the behaviour of commodity prices. *Rev. Econ. Stud.* 59 (1), 1–23.

- Dees, S., Brinca, P.S., 2013. Consumer confidence as a predictor of consumption spending: evidence for the United States and the Euro area. *Int. Econ.* 134, 1–14.
- Dincer, H., Yüksel, S., Şenel, S., 2018. Analyzing the global risks for the financial crisis after the great depression using comparative hybrid hesitant fuzzy decision-making models: policy recommendations for sustainable economic growth. *Sustainability* 10 (9), 3126.
- Dou, Y., Li, Y., Dong, K., Ren, X., 2022. Dynamic linkages between economic policy uncertainty and the carbon futures market: does COVID-19 pandemic matter? *Res. Policy* 75, 102455.
- Dreger, C., Kholodilin, K.A., Ulbricht, D., Fidrmuc, J., 2016. Between the hammer and the anvil: the impact of economic sanctions and oil prices on Russia's ruble. *J. Comp. Econ.* 44 (2), 295–308.
- Erdem, F.P., Ünalmış, İ., 2016. Revisiting super-cycles in commodity prices. *Cent. Bank Rev.* 16 (4), 137–142.
- Fang, L., Chen, B., Yu, H., Xiong, C., 2018. The effect of economic policy uncertainty on the long-run correlation between crude oil and the US stock markets. *Financ. Res. Lett.* 24, 56–63.
- Ferriani, F., Gazzani, A., 2023. The Impact of the War in Ukraine on Energy Prices: Consequences for Firms' Financial Performance. *International Economics*.
- François, O., Laval, G., 2011. Deviance information criteria for model selection in approximate Bayesian computation. *Stat. Appl. Genet. Mol. Biol.* 10 (1).
- Ghabri, F.P., Ayadi, A., Guesmi, K., 2021. Fossil energy and clean energy stock markets under COVID-19 pandemic. *Appl. Econ.* 53 (43), 4962–4974.
- Gil-Alana, L.A., Gupta, R., 2014. Persistence and cycles in historical oil price data. *Energy Econ.* 45, 511–516.
- Gong, X., Guan, K., Chen, L., Liu, T., Fu, C., 2021. What drives oil prices?—a Markov switching VAR approach. *Res. Policy* 74, 102316.
- Goodell, J.W., Gurdgiev, C., Paltrinieri, A., Piserà, S., 2023. Global energy supply risk: evidence from the reactions of European natural gas futures to Nord stream announcements. *Energy Econ.* 106838.
- Gu, X., Zhu, Z., Yu, M., 2021. The macro effects of GPR and EPU indexes over the global oil market—are the two types of uncertainty shock alike? *Energy Econ.* 100, 105394.
- Hemrit, W., Benlagha, N., 2021. Does renewable energy index respond to the pandemic uncertainty? *Renew. Energy* 177, 336–347.
- Huang, H., Liu, H., Yang, B., 2021. Economic policy uncertainty and executive turnover. *China J. Account. Res.* 14 (1), 83–100.
- International Energy Agency, 2021. Net Zero by 2050, IEA, Paris (last accessed 15 June 2022). <https://www.iea.org/reports/net-zero-by-2050>.
- Joo, K., Suh, J.H., Lee, D., Ahn, K., 2020. Impact of the global financial crisis on the crude oil market. *Energy. Strat. Rev.* 30, 100516.
- Kang, W., Ratti, R.A., 2013. Structural oil price shocks and policy uncertainty. *Econ. Model.* 35, 314–319.
- Karkowska, R., Urjasz, S., 2023a. How does the Russian-Ukrainian war change connectedness and hedging opportunities? Comparison between dirty and clean energy markets versus global stock indices. *J. Int. Financ. Mark. Inst. Money* 85, 101768.
- Karkowska, R., Urjasz, S., 2023b. How does the Russian-Ukrainian war change connectedness and hedging opportunities? Comparison between dirty and clean energy markets versus global stock indices. *J. Int. Financ. Mark. Inst. Money* 85, 101768.
- Khan, K., Su, C.W., 2022. Does policy uncertainty threaten renewable energy? Evidence from G7 countries. *Environ. Sci. Pollut. Res.* 29 (23), 34813–34829.
- Khan, K., Su, C.W., Khurshid, A., 2022. Do booms and busts identify bubbles in energy prices? *Res. Policy* 76, 102556.
- Kim, C.J., Nelson, C.R., 1999. Has the US economy become more stable? A Bayesian approach based on a Markov-switching model of the business cycle. *Rev. Econ. Stat.* 81 (4), 608–616.
- Kondratieff, N.D., 1925. The static and the dynamic view of economics. *Q. J. Econ.* 39 (4), 575–583.
- Krolzig, H.M., 2013. Markov-Switching Vector Autoregressions: Modelling, Statistical Inference, and Application to Business Cycle Analysis, vol. 454. Springer Science & Business Media.
- Li, Z., Li, T., 2022. Economic sanctions and regional differences: evidence from sanctions on Russia. *Sustainability* 14 (10), 6112.
- Liu, T., Nakajima, T., Hamori, S., 2021. The impact of economic uncertainty caused by COVID-19 on renewable energy stocks. *Empir. Econ.* 1–21.
- Mahlstein, K., McDaniel, C., Schropp, S., Tsigas, M., 2022. Estimating the economic effects of sanctions on Russia: an allied trade embargo. *World Econ.* 45 (11), 3344–3383.
- Maneejuk, P., Kaewtatip, N., Jaipong, P., Yamaka, W., 2022. The transition of the global financial markets' connectedness during the COVID-19 pandemic. *N. Am. J. Econ. Financ.* 101816.
- Mejdoub, H., Ghorbel, A., 2018. Conditional dependence between oil price and stock prices of renewable energy: a vine copula approach. *Econ. Polit. Stud.* 6 (2), 176–193.
- Mohammed, K.S., Usman, M., Ahmad, P., Bulgamaa, U., 2023. Do all renewable energy stocks react to the war in Ukraine? Russo-Ukrainian conflict perspective. *Environ. Sci. Pollut. Res.* 30 (13), 36782–36793.
- Musah, M., Kong, Y., Mensah, I.A., Antwi, S.K., Donkor, M., 2020. The link between carbon emissions, renewable energy consumption, and economic growth: a heterogeneous panel evidence from West Africa. *Environ. Sci. Pollut. Res.* 27 (23).
- Natorski, M., Pomorska, K., 2017. Trust and decision-making in times of crisis: the EU's response to the events in Ukraine. *JCMS: J. Common Mark. Stud.* 55 (1), 54–70.
- Nerlinger, M., Utz, S., 2022. The impact of the Russia-Ukraine conflict on energy firms: a capital market perspective. *Financ. Res. Lett.* 50, 103243.
- Ott, M., Tatom, J.A., 1986. Are energy prices cyclical? *Energy Econ.* 8 (4), 227–236.
- Paiva, A.S.S., Rivera-Castro, M.A., Andrade, R.F.S., 2018. DCCA analysis of renewable and conventional energy prices. *Phys. A: Stat. Mech. Appl.* 490, 1408–1414.
- Pata, U.K., Kartal, M.T., Zafar, M.W., 2023. Environmental reverberations of geopolitical risk and economic policy uncertainty resulting from the Russia-Ukraine conflict: a wavelet based approach for sectoral CO2 emissions. *Environ. Res.* 231, 116034.
- Pototschnig, A., Glachant, J.M., Meeus, L., Ranci, P., 2022. Recent Energy Price Dynamics and Market Enhancements for the Future Energy Transition. European University Institute.
- Qin, M., Su, C.W., Hao, L.N., Tao, R., 2020. The stability of US economic policy: does it really matter for oil price? *Energy* 198, 117315.
- Reboredo, J.C., Ugolini, A., Chen, Y., 2019. Interdependence between renewable-energy and low-carbon stock prices. *Energies* 12 (23), 4461.
- Regnier, E., 2007. Oil and energy price volatility. *Energy Econ.* 29 (3), 405–427.
- Reinhart, C.M., Wickham, P., 1994. Commodity prices: cyclical weakness or secular decline? *Staff Pap.* 41 (2), 175–213.
- Ringim, S.H., Alhassan, A., Güngör, H., Bekun, F.V., 2022. Economic policy uncertainty and energy prices: empirical evidence from multivariate DCC-GARCH models. *Energies* 15 (10), 3712.
- Saâdaoui, F., Jabeur, S.B., 2023. Analyzing the influence of geopolitical risks on European power prices using a multiresolution causal neural network. *Energy Econ.* 106793.
- Schumacher, D., 1985. The 1973 oil crisis and its aftermath. In: *Energy: Crisis or Opportunity?* Palgrave, London, pp. 21–41.
- Shafiee, S., Topal, E., 2010. A long-term view of worldwide fossil fuel prices. *Appl. Energy* 87 (3), 988–1000.
- Shahrestani, P., Rafiei, M., 2020. The impact of oil price shocks on Tehran stock exchange returns: application of the Markov switching vector autoregressive models. *Res. Policy* 65, 101579.
- Shen, L., Hong, Y., 2023. Can geopolitical risks excite Germany economic policy uncertainty: rethinking in the context of the Russia-Ukraine conflict. *Financ. Res. Lett.* 51, 103420.
- Sims, C.A., Zha, T., 1998. Bayesian methods for dynamic multivariate models. *Int. Econ. Rev.* 949–968.
- Steffen, B., Patt, A., 2022. A historical turning point? Early evidence on how the Russia-Ukraine war changes public support for clean energy policies. *Energy Res. Soc. Sci.* 91, 102758.
- Sun, C., Zhan, Y., Peng, Y., Cai, W., 2022. Crude oil price and exchange rate: evidence from the period before and after the launch of China's crude oil futures. *Energy Econ.* 105, 105707.
- Umar, M., Riaz, Y., Yousaf, I., 2022. Impact of Russian-Ukraine war on clean energy, conventional energy, and metal markets: evidence from event study approach. *Res. Policy* 79, 102966.
- Wang, X., Li, J., Ren, X., Bu, R., Jawadi, F., 2023. Economic policy uncertainty and dynamic correlations in energy markets: assessment and solutions. *Energy Econ.* 117, 106475.
- Yag'ya, V., Yagya, T., 2021. Geography of oil and gas of the Russian federation in the world markets. In: *Proceedings of Topical Issues in International Political Geography*, pp. 236–243. Springer International Publishing.
- Yu, Y., Guo, S., Chang, X., 2022. Oil prices volatility and economic performance during COVID-19 and financial crises of 2007–2008. *Res. Policy* 75, 102531.
- Zhou, X.Y., Lu, G., Xu, Z., Yan, X., Khu, S.T., Yang, J., Zhao, J., 2023. Influence of Russia-Ukraine war on the global energy and food security. *Resour. Conserv. Recycl.* 188, 106657.